

ZACHARY SMITH

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PROFESSIONAL SUMMARY

Data Science graduate from Pennsylvania State University with hands-on experience in healthcare analytics, machine learning, and data visualization. Skilled in Python, SQL, React, Next.js, and modern AI/ML frameworks, with a portfolio of self-directed machine learning projects applying predictive modeling and NLP to gaming data, plus experience transforming complex datasets into actionable insights and automated reporting solutions.

PROFESSIONAL EXPERIENCE

Data Science Intern – Pacific Healthworks | February 2023 – July 2026

- Structured, analyzed, and visualized healthcare data to identify operational trends and actionable insights for hospital efficiency.
- Leveraged MySQL, SSMS, and Tableau for comprehensive data cleaning, transformation, and analysis.
- Presented data-driven findings and recommendations to clinical leadership to inform decision-making processes.
- Automated workflow processes to generate updated data visualizations and reporting tables.
- Collaborated with medical professionals to investigate and resolve data inconsistencies.

Intern – Career Movement | May 2023 – Aug 2023

- Conducted market research on job trends and industry best practices.
- Developed strong communication skills through cross-functional team and client interactions.
- Performed data entry and organizational tasks to maintain data integrity.

Concession Stand Worker – Pennsylvania State University | 2024 – 2025

- Collaborated effectively within team environments to deliver efficient customer service.

EDUCATION

B.S. Data Science – Pennsylvania State University | University Park, PA | 2022 – May 2026

High School Diploma – Mira Costa High School | 2018 – 2022

SKILLS

Languages: Python, R, SQL

Web & Frameworks: React, Next.js

AI Frameworks: TensorFlow, PyTorch, scikit-learn, HuggingFace Transformers, FAISS

Tools: MySQL, SSMS, Tableau, VSCode, Chart.js

Core Competencies: Machine Learning, Data Analysis, Statistical Analysis, Data Visualization, Workflow Automation, NLP & Embeddings

PROJECTS

Game Recommendation LLM | Python, HuggingFace, Transformers, SQL, FAISS

- Built a recommendation engine that matches players with games they'll enjoy using semantic embeddings of game descriptions generated with HuggingFace transformers.
- Designed a FAISS vector index to run fast, content-based similarity queries at scale.

Steam Game Success Predictor | Python, Steam API, XGBoost, NLP/VADER, Matplotlib

- Built an end-to-end ML pipeline using XGBoost to predict whether a game will succeed, achieving 85.7% classification accuracy.
- Pulled live telemetry and metadata from the SteamSpy API (concurrent users, pricing, developer history, tags) and ran VADER sentiment analysis on game descriptions as model features.

Player Churn Prediction | Python, XGBoost, Scikit-Learn, Pandas, Data Viz

- Developed a survival analysis and classification model to flag live-service players at risk of churning based on play frequency, session length, and progression telemetry.
- Enabled proactive retention strategy design by surfacing churn risk scores ahead of player drop-off.

BetTruth: Sportsbook Analytics & Odds Simulator | JavaScript (ES6+), Chart.js, HTML5/CSS3, Monte Carlo Sim

- Built an interactive web app and Monte Carlo bankroll simulator modeling sportsbook overround (vig), parlay compounding, and expected value.
- Implemented real-time statistical modeling and dynamic Chart.js visualizations to demonstrate risk simulation results.